Genotype and Phenotype Lab

Week Three

1. Purpose and Objectives

The purpose of this investigation is to determine students' genotype for a taste receptor gene and correlate the genotype with the phenotype observed.

After completing this lab, you should be able to:

- Understand the difference between phenotype and genotype.
- Describe how DNA (deoxyribonucleic acid) sequence (genotype) determines an organism's characteristics (phenotype)
- Observe how a change in DNA sequence may alter phenotype
- Explain how polymerase chain reaction (PCR) can be used to observe differences in genotype
- Describe how DNA gel electrophoresis works.
- Use a bioinformatics resource tool to obtain and explore gene sequences.

Prelab

- Review PCR and the role of primers.
- Read (and watch videos) from https://www.visiblebody.com/blog/dna-and-rna-basics-replication-transcription-and-translation
- Read about Transcription and Translation, Chapter 15, Genes and Proteins, OpenStax Biology 2e

Background

Bioinformatics is a science that uses statistical methods and computers to analyze biological data. It can be applied to a variety of fields ranging from ecology to chemistry. A common application is analysis of nucleotide and amino acid sequence data. As DNA sequencing has become relatively easy over the last couple of decades an enormous amount of sequence data has been generated. This data is deposited in databases and analyzed using computational methods. One obvious task is to compare sequences to one another to determine how similar one gene is to another or to compare one organism to another. This lab will use a sequence database called "genbank" that is part of the National Center for Biotechnology Information (NCBI) and has been developed by the US National Institutes of Health (NIH). Scientists around the world upload sequence data from their experiments to this database. The NIH and the National Library of Medicine also provide a commonly used

analysis tool called the Basic Local Alignment Search Tool (BLAST) that we will use to compare nucleotide and amino acid sequences of Coronavirus spike protein genes.

Prelab Questions

1. Explain why PCR only copies a segment of the gene from the template, ie why is the band on the gel smaller than the TAS2R38 gene, which is more than 1000 base pairs in length?
2. The process of transcription makes a/an _____ copy of _____.
3. What type of molecule is made in translation? How is the amino acid sequence determined?
4. The genes that an organism has are its _____. The organism's observable characteristics are its _____.
5. The genotype characteristic that was examined in this lab is whether a single nucleotide polymorphism is present in the bitter taste receptor gene TAS2R38. How does the genotype for TAS2R38 affect phenotype?

Procedure DNA Database Search

1. In the first step you will use the primer sequences to search for matching sequences in the NCBI database. Note that the primer sequences can be found on p S-11 of the handout. If you pull up the handout pages from Canvas you should be able to copy and paste these.
2. Follow the procedure on pp. S-11 to S-12 for step 1 parts a thru i.
3. The descriptions page that comes up lists entries into the database that the Blast program found matched the sequence that you entered in the "query" box. These are listed as "sequences producing significant alignments" and are sometimes referred to as "hits." **What sequence(s) did you enter into the query box when you did your blast search?**
4. **What human (Homo sapiens) gene is at the top of the list of sequences producing significant alignments?**
5. **What other organisms' genes appear on the list as well? Name two.**
6. There is a great deal of information included on the Descriptions page. The "Query Cover" is the percentage of your query sequence that is found in each sequence entry listed. It is only 98% for the first entries listed because you entered an "N" between the primer sequences. The "Percent Identity" is the percentage of nucleotides in the sequence entry that match the nucleotides in the query sequence. **What is the percent identity (3rd column from the right) for the first ten entries?**

7. Each of the sequences listed on the description page was entered into the database by a group of researchers that have isolated and sequenced that gene in the laboratory. Each uploaded sequence is given a unique identifier called an accession number. These are listed in the far-right column of the descriptions page.
8. Find the entry that says Homo sapiens PTC bitter taste receptor (PTC) gene, PTC taster allele, accession number AY258597.1. This entry is described as "complete cds." Cds stands for coding sequence so this entry contains the nucleotide sequence for the part of this gene that determines the amino acid sequence of the taste receptor protein. Next to the accession number column is the accession length. This is the number of nucleotides in the sequence entry. **What is the length of the nucleotide coding sequence for this gene?**
9. Click on this entry. This will take you to a page that shows the alignment of your query sequence and the database entry (subject). Matching nucleotides are indicated by vertical dashes between the query sequence and subject. The first place in the gene where the sequence matches your query is for one of the primers that was used in the PCR procedure. **At what nucleotides in the coding sequence for this gene does the primer match the coding sequence? What nucleotides in the coding sequence match the other primer?**
10. Your PCR product contains both primer sequences and the region of the TAS2R38 gene in between the primers. Subtract the lowest nucleotide position for the primer match from the highest nucleotide position. Add one to get the PCR product size since the product includes both ends. **What size (in base pairs) should your PCR product be?**
11. Scroll up to the PTC non-taster allele entry. **Are there any differences in the alignment position or percent identity compared to the taster allele? Should the PCR product be the same size as for the taster allele?**

Procedure Find and Copy the DNA sequence for the Human PTC Taster Gene

1. On either the alignment page or the description page click on the accession number for the PTC-taster allele (labeled "Sequence ID" on the alignments page). This will bring up more information about the database entry including the journal reference where the sequence is published and the authors of the papers.
2. Scroll to the bottom of the page to find the nucleotide sequence of the database entry. **Copy and paste this sequence into a Word document. Label it TAS2R38 taster allele cds. Print this document.**
3. In the middle of the page are listed four variations that occur in the sequence of this gene. Three of these differ between the taster allele and the non-taster allele. Click on

the word "variation" for the first of these, at position 145. This will highlight the base at this position in the sequence below. **Circle this base in your printout. What base is found at this position in the taster allele? What does the variation entry say it is replaced with in the non-taster allele (labeled replace =)?**

4. **Find and underline the bases that are included in the PCR primers on your printout.**

Procedure: RNA Sequence and Translation

1. The database entry that your blast search found lists the coding sequence in DNA nucleotides. This is how sequences are entered into databases so that they can be searched. However, the coding sequence does not contain the sequence of the entire TAS2R38 gene but only the nucleotides of the coding strand for parts of the gene that get transcribed and included in mature mRNA. Another way to say this is that the entry is the DNA version of the nucleotides found in mRNA. To translate the nucleotide sequence into amino acid sequence we need to convert it into RNA nucleotides. **What nucleotide differs between DNA and RNA?**
2. All transcripts start with the same codon (three base pair sequence). **Rewrite the first three bases of the coding sequence as RNA nucleotides.** This is the first codon to be translated into amino acid sequence. Look at a chart of the genetic code. There is one posted on your Canvas page. **What amino acid will be the first amino acid of the protein?**
3. **Rewrite bases 121 to 150 as RNA sequence.** Note that this is *not* transcription. You are simply changing one of the nucleotides to an RNA nucleotide.
4. The non-taster allele has a single nucleotide polymorphism at position 145, which you circled earlier. The cytosine (c) at this position is replaced by a guanine (g). **For what amino acid does the non-taster allele codon from nucleotides 145 to 147 code? What amino acid was at this position in the taster allele?**

Procedure: Restriction Site Polymorphism

1. Recall that when you entered the primer sequences into the Blast search you changed the second to last "G" in the forward primer to an "A."
2. **Write the DNA nucleotide sequence for the taster allele that you printed out above from nucleotide 141 to nucleotide 150.**
3. **Write the last four nucleotides of the primer sequence below the nucleotides copied in step two.** Although the primer sequence differs at one nucleotide from the gene sequence in your DNA, it will bind well enough to the TAS2R38 to make a PCR product. All of the resulting copies will have the "G" in the PCR product rather than the "A" from the gene sequence. This is done to create a site where the restriction enzyme HaeIII can cut the PCR product.

4. **Write the nucleotide sequence that will be in the PCR product starting with the last four nucleotides of the primer sequence and ending 6 nucleotides past the end of the primer.**
5. **The HaeIII enzyme cuts DNA where it finds the nucleotide sequence "ggcc". Does this sequence appear in the PCR product from the taster allele?**
6. **Write the nucleotide sequence for this part of the PCR product for the non-taster allele. Can the HaeIII enzyme cut the PCR product from the non-taster allele?**

Procedure: Restriction Site Polymorphism

1. **Complete the Results and Discussion section found on pp. S-23 to S-25 of the handout.**